State-of-Charge Occupancy and Battery Aging in Energy–Reserve Bidding Under Price Stress: A Diagnostic Using German Market Data

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Abstract—Joint energy and reserve offers can require a battery energy storage system (BESS) to retain energy for upward activation. Throughput and depth-of-discharge (DoD) costs do not directly price this retained energy, whereas calendar aging depends in part on state of charge (SoC). We study when an SoC-occupancy term changes an optimized energy–reserve position in a virtual power plant (VPP). A daily mean–CVaR model represents a 2.5 MW/5 MWh BESS with throughput, daily-depth, and layered calendar–cycle (ReLCCA) aging costs. The input panel contains 262 complete days of German market and system data from December 2025 through August 2026; reserve-capacity and imbalance settlements are modeled using proxies. In the originally reported runs, throughput and Xu formulations select a long energy-only position on 1 May 2026, when the day-ahead price reaches a reported −499 EUR/MWh; ReLCCA remains short. The reported wholesale-profit difference is EUR 1,312, largely associated with a EUR 1,246 difference in day-ahead energy revenue, while direct aging cost differs by EUR 4. A reconstruction using the archived panel and available code does not reproduce this ranking: P1−B2 is negative for all three aging laws in both relaxed-LP and charge/discharge-exclusive MILP runs. The original numerical ranking therefore remains unverified without the original model and parameter set.

Index Terms—battery energy storage, battery degradation, calendar aging, CVaR, energy–reserve bidding, linear programming, SMARD, virtual power plant.

I. INTRODUCTION

Virtual power plants (VPPs) aggregate distributed energy resources and expose a heterogeneous portfolio to electricity markets through coordinated scheduling [1]–[4]. Their short-term value depends on the interaction between uncertain renewable production, load, market prices, storage flexibility, and ancillary-service commitments. This has motivated stochastic, robust, and risk-aware bidding models for VPPs and storage resources [5]–[8].

For BESSs, however, the economic value of flexibility is inseparable from degradation. Battery aging is commonly separated into cycle aging and calendar aging. Experimental and semi-empirical studies have shown that degradation depends on operating depth, state of charge, temperature, time, and electrochemical conditions [9]–[14]. Market-oriented optimization has consequently adopted throughput costs, cycle-depth functions, and more detailed aging representations to maintain computational tractability [15]–[20].

A specific modeling issue arises when the same BESS is used for energy arbitrage and reserve capacity. An upward reserve commitment requires sufficient energy to remain available for deployment. Thus, some stored energy is deliberately parked rather than cycled. If calendar aging depends on SoC, the economic exposure of this parked energy is not represented by a pure throughput penalty. The resulting effect is potentially indirect: a small change in the degradation term can alter the marginal value of stored energy and, through the optimization constraints, change the preferred market position.

The objective is not to claim that calendar-aware degradation is universally superior to cycle-centric formulations. Instead, the paper asks a narrower question: under what price and residual-portfolio conditions does explicit pricing of SoC occupancy change an optimized mean–CVaR energy–reserve position? The question is examined through paired comparisons in which the plant, scenario inputs, market assumptions, risk parameters, and solver are held fixed while the aging representation is changed. The analysis is a model diagnostic, not a population-wide test or validation of a cell-aging law.

The contributions are fourfold. First, a common linear market formulation allows three aging representations to be compared without changing the surrounding optimization model. Second, the ReLCCA (layered calendar–cycle aging) representation combines SoC-occupancy and band-flow costs with an explicit upward-reserve energy-parking constraint. Third, a profit-decomposition procedure separates direct aging expenditure from changes in the optimized day-ahead energy position. Fourth, a controlled negative-price intervention and a panel-level scan of all twenty qualifying negative-price days — an out-of-sample-style check that does not rely on the four isolation dates alone — examine whether price depth alone is sufficient to

explain the reported reversal. These contributions concern model structure and diagnostic evidence; they do not establish universal superiority of one aging law.

II. RELATED WORK AND RESEARCH GAP

The literature relevant to this study can be organized into four strands: VPP market participation, uncertainty and risk, battery degradation, and energy–reserve co-optimization. Foundational VPP studies established aggregation and coordinated market participation as an operational paradigm [1]–[4]. Stochastic and robust formulations subsequently addressed uncertainty in renewable production and prices, including price-taker and adaptive-robust offering models [5]–[7]. Risk-averse energy and reserve scheduling has also been studied explicitly for VPPs [8].

A second strand concerns battery aging. Foundational reviews and experiments distinguish calendar and cycle mechanisms and document the dependence of calendar aging on SoC [9]–[14]. Optimization studies translate these relationships into computational models. Hesse et al. use mixed-integer linear optimization for aging- and efficiency-aware arbitrage [15]. Xu et al. develop market-oriented cycle-aging cost formulations [16], [17], while Shi et al. analyze cycle-aging mechanisms in pay-for-performance operation [18]. More recent work considers convex-hull and electro-thermal representations to balance aging fidelity and optimization complexity [19], [20].

A third strand concerns reserve participation. Storage has been co-optimized across energy and reserve products, including formulations with probabilistic delivery guarantees and German secondary-control-reserve applications [21]–[24]. These studies establish that reserve commitments impose operational and opportunity-cost constraints. Related wind–battery VPP models incorporate degradation and reserve scheduling jointly [24], [25].

A fourth strand is risk-aware bidding. CVaR provides a tractable representation of lower-tail risk [26]–[28] and has been applied to VPP, PV-storage, and energy–frequency-regulation bidding [8], [29]–[31]. Distributionally robust VPP models further extend the treatment of uncertainty [32].

The gap addressed here is narrower than any of these individual topics. The paper does not claim novelty in degradation, CVaR, stochastic bidding, or reserve participation separately. Rather, it isolates the incremental effect of making SoC occupancy economically visible while retaining cycle-related costs, upward-reserve parking, and a common mean–CVaR market model. The empirical component is framed as a diagnostic because reserve-capacity and imbalance series in the study package are proxies rather than day-specific German settlement records.

TABLE I. Comparison of representative studies and positioning of the present work

Study	Aging	Risk / uncertainty	Energy–reserve	Relevance
Hesse et al. [15]	Aging + efficiency	Deterministic	Energy arbitrage	Optimization-aware degradation
Xu et al. [16], [17]	Cycle depth	Market uncertainty	Energy / reserve	Market-oriented cycle cost
Shi et al. [18]	Cycle aging	Operational uncertainty	Regulation	Cycle-aging control
Toubeau et al. [21]	—	Probabilistic	Energy / reserve	Reserve delivery
Merten et al. [22], [23]	Battery-aware	Market forecasting	aFRR	German reserve participation
Fernández-Muñoz and Pérez-Díaz [24]	DoD / rate	Stochastic	DA + secondary reserve	Wind–battery VPP
Ghanaee et al. [25]	Calendar + cycle	Deterministic	DA + reserve	Recent degradation benchmark
Wang et al. [31]	Operational constraints	CVaR	Energy + regulation	Risk-aware benchmark

This work	Occupancy + cycle	Mean–CVaR	DA + aFRR capacity	Controlled diagnosis of bid changes

Among the studies summarized in Table I, [25] combines calendar and cycle aging with day-ahead energy–reserve scheduling, but uses deterministic inputs. The present comparison examines the additional effect of SoC-dependent aging in a mean–CVaR bid. This positioning is limited to the studies reviewed here and should not be read as an exhaustive claim about the entire literature.

III. SYSTEM, DATA, AND INFORMATION STRUCTURE

A. VPP configuration

The VPP is modeled as a price-taking portfolio in the German DE-LU bidding area. It contains a 5 MW-class industrial load, 4 MW photovoltaic generation, 2 MW wind generation, a 2.5 MW/5 MWh stationary BESS, and a 1.8 MW dispatchable generator. The BESS has 0.95 charging and discharging efficiencies and operates within a 10–90% SoC range. A cyclic daily boundary condition is imposed. The generator has a marginal operating cost of 68 EUR/MWh and a 1.2 MW/h ramp limit. The default aggregate reserve cap is 1.5 MW. These values are fixed study parameters and are not fitted to the reported outcomes.

TABLE II. VPP, battery, market, and risk parameters used in the numerical study

Parameter	Value	Role
BESS power / energy	2.50 MW / 5.00 MWh	Study plant
PV / wind / load class	4 / 2 / 5 MW	Study plant
η_c / η_d	0.95 / 0.95	Efficiency
SoC range	10–90%	Operational constraint
DG capacity / cost	1.8 MW / 68 EUR/MWh	Dispatchable resource
DG ramp	1.2 MW/h	Operational constraint
Reserve cap	1.5 MW	Default sensitivity parameter
CVaR confidence α	0.95	Tail confidence
CVaR weight β	0.05	Risk weight
SoC bands N	8	Default ReLCCA resolution
Solver	HiGHS	Open-source LP solver
Random seed	20260821	Reproducibility

Table II lists the parameters specified for the reported study. The manuscript does not state the reserve-sustainment duration τ, initial SoC, or band-specific cycle coefficients γ used by the available reconstruction; those defaults are not treated as original study values.

B. Data and scaling

The panel covers 1 December 2025 00:00 through 21 August 2026 07:00 Berlin time. It contains 6,319 hourly rows across 264 calendar dates: 262 complete days, the 23-hour daylight-saving day on 29 March, and the first 8 hours of 21 August. Day-ahead prices for DE-LU are sourced through the Energy-Charts API from Bundesnetzagentur | SMARD.de; German load, solar, and wind series are from Fraunhofer ISE Energy-Charts [33], [34]. The API's 15-minute observations are averaged to local-time hourly values. Load is scaled so its study-period 95th percentile maps to 5 MW; the study-window solar maximum maps to 4 MW, and the combined onshore/offshore wind maximum maps to 2 MW. Price forecasts use the mean of 48- and

168-hour lags; load, PV, and wind use 48-hour persistence. Daily forecast-error vectors are block-bootstrapped from complete days no later than D−2, with uniform scenario weights. Raw responses, the processed panel, retrieval timestamp, licenses, URLs, scaling factors, and SHA-256 hashes are archived with the data package. These are national system series, not battery telemetry.

The information structure separates first-stage market commitments from scenario-dependent operating recourse. Day-ahead energy and reserve positions are common across scenarios, whereas charging, discharging, generator output, imbalance, and band-level operating variables respond to each scenario. Scenario probabilities are uniform within the daily scenario set.

C. Reserve and imbalance representation

The reserve-capacity inputs are six positive- and negative-direction German aFRR four-hour block means reported by Modo Energy and Pexapark [35], [36]: positive values are 4.5, 12.0, 47.1, 18.4, 35.6, and 8.0 EUR/MW/h; negative values are 6.0, 15.0, 36.9, 57.4, 20.0, and 2.4 EUR/MW/h. They are held fixed across operating days and serve as capacity-price proxies, not daily auction results.

The imbalance component is represented by a documented two-price proxy based on the day-ahead price. For a short deviation, the settlement proxy is λDA + 18 + 0.12|λDA|; for a long deviation it is λDA − 10 − 0.08|λDA|, subject to minimum/maximum spreads of +5 and −3 EUR/MWh, respectively. A 150 EUR/MWh clipping case is evaluated only as a robustness test. Because directly mounted day-specific reBAP data were not available in the study package, the reported wholesale results are **simulated profits under these settlement assumptions**, not realized German settlement profits.

IV. MATHEMATICAL FORMULATION

A. Common market model

All three aging laws are embedded in the same daily optimization problem. Let t denote hour, ω scenario, and ℓ SoC band. The first-stage variables are the day-ahead energy position p_t^DA and reserve commitments r_t↑ and r_t↓. Recourse variables include battery charging p_tω^ch, discharging p_tω^dis, stored energy E_tω, generator output g_tω, net imbalance z_tω, and ReLCCA band variables.

The physical balance for each hour and scenario is written as

$-P_t^{DA} + z_{t,\omega} + P_{t,\omega}^{dis} - P_{t,\omega}^{ch} + g_{t,\omega} + W_{t,\omega} + PV_{t,\omega} = L_{t,\omega}$	(1)

A positive p_t^DA denotes a day-ahead sale, and a positive z_t,ω denotes energy purchased in the imbalance market. The balance therefore subtracts the day-ahead sale and adds the imbalance purchase. At a negative price, a positive purchase is a long position. The implementation uses separate nonnegative short and long imbalance variables; z represents their signed net position. Renewable and generator limits are linear, and generator ramping is represented by intertemporal inequalities.$P_t^{DA} z_{t,\omega}$

B. Battery dynamics and reserve feasibility

$E_{t,\omega} = E_{t-1,\omega} + \eta_c P_{t,\omega}^{ch} \Delta t - \frac{P_{t,\omega}^{dis}}{\eta_d} \Delta t$	(2)
$E_{min} \le E_{t,\omega} \le E_{max}, 0 \le P_{t,\omega}^{ch} \le P_B, 0 \le P_{t,\omega}^{dis} \le P_B$	(3)
E_{t,ω} = E_min + Σ_ℓ ΔE_ℓ u_{t,ω,ℓ}, 0 ≤ u_{t,ω,ℓ} ≤ u_{t,ω,ℓ−1} ≤ 1	(3a)
E_(0,ω)=E_(T,ω)	(4)

The cyclic condition is applied separately to each scenario. Upward reserve is constrained by both power capability and an energy-parking requirement. A generic form consistent with the implemented model is

$r_t^{\uparrow} \le \bar{P}_B - P_{t,\omega}^{dis}$	(5)
$E_{t,\omega} \ge E_{min} + \frac{\tau^{\uparrow} r_t^{\uparrow}}{\eta_d}$	(6)
r_t^↓ ≤ P_B − p^ch_{t,ω}, E_{t,ω} + η_c τ^↓ r_t^↓ ≤ E_max	(6a)

where τ↑ is the fixed reserve-sustainment parameter used by the study. The constraint prices the energy that must remain available for upward deployment; it does not simulate the detailed activation trajectory or mileage of an aFRR signal.

The baseline LP omits binary charge/discharge exclusivity, so simultaneous charging and discharging are feasible. The economic terms do not guarantee that this overlap is zero; we therefore report an overlap diagnostic and a binary-exclusive MILP sensitivity case. LP results apply only to the relaxed formulation. In the accompanying reconstruction's relaxed-LP runs for 1 May 2026, hourly simultaneous charge–discharge reaches 11.99 MWh under the Xu law (mean 6.44 MWh, present in 87% of scenarios) and up to 9.74 MWh under the other laws, whereas the binary-exclusive MILP re-solves produce overlap of order 10^{-14} MWh, i.e., numerical noise, for all three aging laws. The LP dispatch therefore overstates usable arbitrage flexibility at negative prices, and LP outputs are reported only as relaxation checks, never as final results.

C. Three aging representations

1) Throughput model. The benchmark assigns a constant marginal aging cost to charge and discharge:

$C_{t,\omega}^{THR} = c_{thr}\left(P_{t,\omega}^{dis} + P_{t,\omega}^{ch}\right)\Delta t$, $c_{thr} = 1.5$ EUR/MWh	(7)

Idle stored energy is not directly priced by this term.

2) Daily DoD model. The Xu benchmark represents daily depth by the normalized range of stored energy. Let Emax,d and Emin,d be the maximum and minimum daily stored energy. The daily depth variable d_d satisfies

$d_d \ge \frac{E_{max,d} - E_{min,d}}{E_B}$	(8)

and its cost is represented by a convex piecewise-linear epigraph with segment widths 0.20, 0.30, 0.30, and 0.20 and slopes 10, 40, 80, and 120 EUR per unit depth [16].

$d_d = \sum_{k=1}^{4} \delta_{d,k}$	(9)
$C_d^{DoD} = \sum_{k=1}^{4} m_k \delta_{d,k}$	(10)
$0 \le \delta_{d,k} \le w_k$, $k = 1,..,4$, $w = [0.2,0.3,0.3,0.2], m = [10,40,80,120]$	(11)

3) ReLCCA. The usable interval $[E_{min}, E_{max}]$ is divided into N equal-width SoC bands. Occupancy variables $u_{t,\omega,l}$ are represented cumulatively so that band filling is monotone. Band-level charge and discharge flows connect occupancy changes to battery movement. The ReLCCA aging cost is

$C_{t,\omega}^{\mathrm{ReLCCA}} = \sum_{\ell} k_{\ell}\, u_{t,\omega,l} \Delta E_l \Delta t + \sum_{l} \gamma_l^{ch} P_{t,\omega,l}^{ch}\, \Delta t + \sum_{l} \gamma_l^{dis} P_{t,\omega,l}^{dis}\, \Delta t$	(12)
$k_{\ell} = \begin{cases} 0.247 & \ell \epsilon L_{low} \\ 0.493 & \ell \epsilon L_{mid} \\ 0.822 & \ell \epsilon L_{high} \end{cases}$ [EUR/MWh/h]	(13)

The first term is the defining occupancy component: energy retained in a high-SoC band incurs a calendar-aging cost even when no energy is cycled during that hour. The second term retains cycle-related costs. All coefficients are fixed parameters, so the formulation remains affine. Each κ comes from one formula: κ = (fade over ten months × 12/10 × replacement cost) / 8,760 h. For the mid plateau this is 0.024 × 1.2 × 150,000 / 8,760 = 0.493 EUR/MWh/h.

The default eight-band κ map uses three plateaus derived from the supplied Keil-based calibration: 0.247, 0.493, and 0.822 EUR/MWh/h for low-, mid-, and high-SoC bands, respectively [12]. These are economic inputs based on published calendar-aging relationships, not a fitted cell-aging model.

D. Mean–CVaR objective

For scenario ω, wholesale profit is represented as

$\Pi_{\omega} = R_{DA,\omega} + R_{aFRR,\omega} - C_{imb,\omega} - C_{DG,\omega} - C_{age,\omega}$	(14)

A retail load-service overlay, when reported, is added after the wholesale optimization and therefore cannot change the market-decision ranking. Formally, the reported profit equals the wholesale objective plus this post-optimization overlay term, which is computed from the realized dispatch schedule. It therefore shifts profit levels without entering the optimization itself; the inter-model ranking in Tables III–VI is driven by the wholesale objective, which is evaluated under identical prices, scenarios, and constraints for all three aging laws, and the overlay is added to each law under exactly the same rule.

Let $L_{\omega} = -\Pi_{\omega}$ denote scenario loss. With confidence level α = 0.95, the Rockafellar–Uryasev representation introduces a threshold ζ and nonnegative excess variables s_{ω}:

$s_{\omega} \geq L_{\omega} - \zeta \ , s_{\omega} \geq 0$	(15)
$CVaR_{\alpha}(L) = \zeta + \frac{1}{1-\alpha} \sum_{\omega} \pi_{\omega} s_{\omega} \ , \alpha = 0.95$	(16)

The implemented objective maximizes expected profit while penalizing lower-tail loss with CVaR risk weight β = 0.05 (Table II):

$max \sum_{\omega} \pi_{\omega} \Pi_{\omega} - \beta CVaR_{\alpha}(L) \ , \beta = 0.05$	(17)

Without binary charge/discharge exclusivity, the objective and constraints are affine and the baseline problem is an LP. The exclusive sensitivity case adds binary variables and is a MILP. The daily DoD epigraph and CVaR use auxiliary variables and linear constraints.

V. EXPERIMENTAL DESIGN

A. Paired identification principle

B2 is the energy-only bid; P1 jointly offers day-ahead energy and aFRR capacity. P1−B2 is the difference in simulated wholesale profit after settlement. For each day, bids are optimized over the same scenario set, then first-stage positions are fixed while recourse is re-optimized on the observed day. Plant, prices, risk settings, and solver are held constant within each aging-law comparison.

B. Isolation days

Four isolation dates are used to represent ordinary and stressed conditions: 20 January, 5 April, 1 May, and 19 August. The default scenario count is K = 30. These dates are diagnostic examples rather than a representative random sample.

C. Counterfactual negative-price intervention

To test whether negative-price depth alone is sufficient to reproduce the observed reversal, the 19 August residual portfolio is frozen while the selected midday price interval is replaced by scaled copies of the 1 May trough. The injected minimum is varied across −50, −80, −200, −300, −414, and −499 EUR/MWh. The experiment therefore changes price severity while preserving the underlying residual structure; it is a counterfactual intervention, not a market forecast.

D. Historical negative-price scan

We scan days with a minimum day-ahead price below −20 EUR/MWh. The 20 qualifying days form an in-sample event set, not a random sample; the scan is descriptive and does not estimate a population probability.

E. Profit decomposition and robustness

Whenever aging formulations produce materially different market positions, wholesale profit is decomposed into day-ahead energy, reserve, imbalance, generator, and aging components. Robustness checks vary the number of SoC bands, calendar coefficient mapping, calendar/cycle components, scenario count, calendar annualization, imbalance clipping, reserve cap, and battery energy capacity. These tests evaluate sign stability within the model and do not constitute electrochemical validation.

VI. RESULTS

The numerical values in Tables III–VI and Figs. 1–3 are the values reported by the original manuscript runs. They are retained here for transparent documentation, but have not been independently reproduced by the available reconstruction. Accordingly, they are reported results rather than verified outputs of the archived reconstruction. The reconstruction discrepancy is stated explicitly wherever it affects interpretation.

TABLE III. Paired isolation-day results under the three battery-aging representations.

K = 30, 180 EUR/MWh load-service overlay included (Differences are computed from unrounded values)

Day	Law	B2 (EUR)	P1 (EUR)	P1−B2 (EUR)
20 Jan	Throughput / Xu / ReLCCA	10,330 / 10,332 / 9,886	11,260 / 11,258 / 11,100	+930 / +926 / +1,214
5 Apr	Throughput / Xu / ReLCCA	15,475 / 15,475 / 15,215	16,676 / 16,676 / 16,603	+1,201 / +1,201 / +1,389

1 May	Throughput / Xu / ReLCCA	14,492 / 14,492 / 13,180	14,246 / 14,225 / 13,696	−246 / −266 / +515	
19 Aug	Throughput / Xu / ReLCCA	10,961 / 10,961 / 10,766	12,268 / 12,268 / 12,245	+1,307 / +1,306 / +1,480	

A. Paired isolation-day results

For the three non-event dates, the original manuscript reports positive P1−B2 values under all three aging laws. For 1 May, it reports a negative differential for throughput and Xu but a positive differential for ReLCCA. The available reconstruction does not reproduce this ranking; the original table values therefore remain unverified. Table III reports isolation-day results with K = 30 and the 180 EUR/MWh load-service overlay, whereas Table V reports the historical negative-price scan with K = 20 and no overlay; values for the same calendar day may therefore differ between the two tables.

B. Controlled price-trough experiment

In the original manuscript's 19 August portfolio intervention, P1−B2 remains positive across the injected troughs: throughput declines from +1,145 to +790 EUR and ReLCCA from +1,353 to +1,056 EUR. Deepening the trough alone did not reproduce the 1 May reversal in this frozen portfolio. These are original reported values, not independently reproduced outputs; the counterfactual is neither a forecast nor a general causal test.

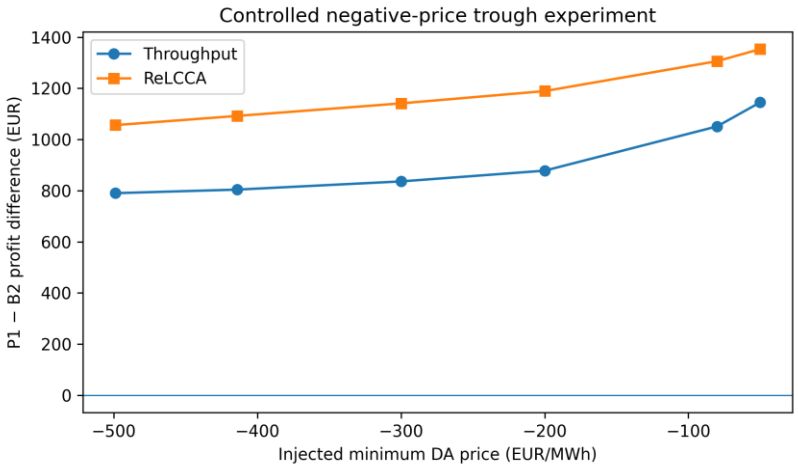


Fig. 1. Controlled negative-price trough experiment using the 19 August residual structure. The intervention varies only the injected midday price minimum; it is not a forecast.

C. 1 May profit decomposition

TABLE IV. 1 May 2026 wholesale profit by component, K = 30, no overlay. B2 = energy-only, P1 = energy + reserve

Model	DA energy	aFRR	Imbalance	DG	Aging	Wholesale
B2–Throughput	+978	0	+600	−1,714	−58	−194
P1–Throughput	−541	+1,333	+436	−1,635	−32	−440
B2–ReLCCA	−268	0	+462	−1,647	−54	−1,506
P1–ReLCCA	−1,238	+1,457	+474	−1,635	−49	−991

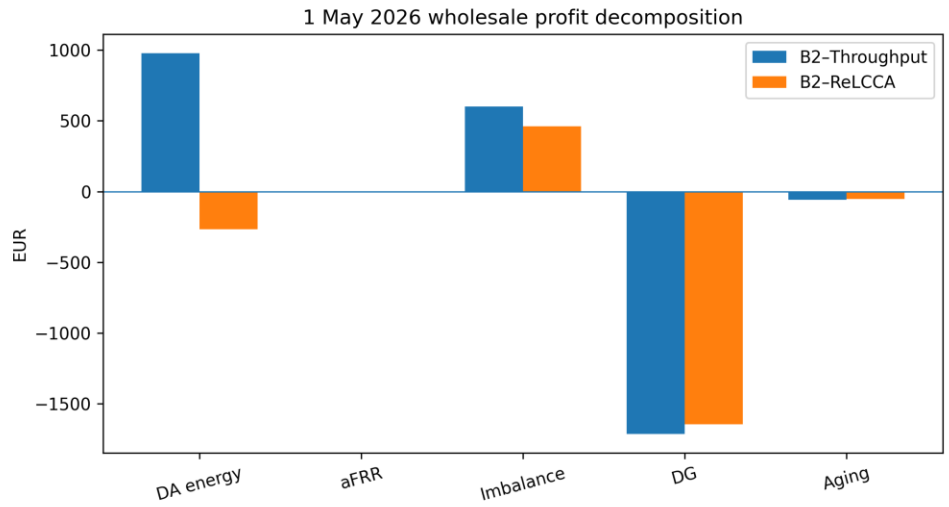


Fig. 2. 1 May 2026 wholesale profit decomposition for the energy-only bid (B2) under the throughput and ReLCCA aging representations

The original manuscript's decomposition reports that B2–Throughput and B2–ReLCCA differ by EUR 1,246 in day-ahead energy revenue (+978 versus −268 EUR) but by only EUR 4 in direct aging cost (−58 versus −54 EUR). Their reported wholesale profits differ by EUR 1,312 (−194 versus −1,506 EUR). If reproduced, this decomposition would locate most of the difference in the optimized market position rather than in direct aging expenditure. The available reconstruction does not reproduce these values, so this mechanism remains a hypothesis supported by the original reported decomposition, not a verified result.

D. Historical negative-price scan

TABLE V. Historical negative-price days below −50 EUR/MWh and corresponding model differentials

Day	Minimum DA price	Throughput P1−B2	Xu P1−B2	ReLCCA P1−B2
1 May	−499	−246	−251	+531
26 Apr	−414	+139	+137	+869
25 Apr	−191	+812	+850	+1,114
2 May	−130	+887	+887	+1,188
6 Apr	−126	+1,043	+1,043	+1,382
5 Apr	−105	+1,182	+1,182	+1,381
11 Apr	−80	+1,100	+1,100	+1,246
24 May	−76	+1,115	+1,115	+1,316
29 Apr	−60	+1,074	+1,060	+1,273

The original manuscript's scan reports one throughput reversal among 20 days below −20 EUR/MWh: 1 May. Table V lists the nine days with minimum day-ahead prices below −50 EUR/MWh. It reports ReLCCA as positive on all 20 days. These are in-sample counts over days with different portfolio states, not probability estimates. The available reconstruction produces two reversals for each aging law; the original scan result is therefore unverified.

E. Numerical and parameter robustness

TABLE VI. Robustness and sensitivity checks for the ReLCCA P1−B2 differential

Check	ReLCCA P1−B2 (EUR)
N = 4 / 6 / 8 / 12, K = 20	+565 / +533 / +576 / +561
κ step / all-mid / all-high / 100% top, K = 30	+515 / +514 / +512 / +515
Calendar only / cycle only / both, K = 30	+587 / +512 / +515
K = 20 / K = 30	+576 / +515
√t annualization (κ × 0.913)	+516
Imbalance |DA| clipped at 150	+541 (throughput −232)
rmax = 0.5 / 2.5 MW, K = 20	+234 / +800 (throughput −91 / −262)
Eb = 2 MWh, K = 20	+409 (throughput +19)

The original manuscript's component checks report P1−B2 values of +587 EUR for calendar-only, +512 EUR for cycle-only, and +515 EUR for the combined model. These values would suggest that SoC dependence, rather than the calendar term alone, drives the reported distinction; however, the checks have not been independently reproduced.

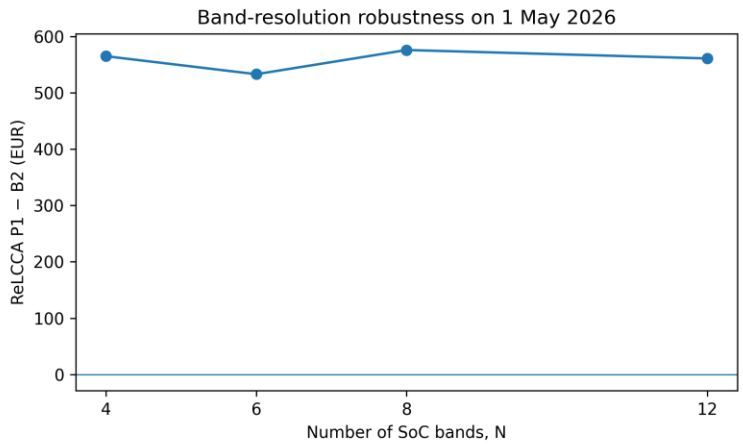


Fig. 3. ReLCCA differential across SoC-band resolutions on 1 May 2026

The original manuscript's sensitivity runs report a ReLCCA P1−B2 gap of +234 to +800 EUR across tested reserve caps and positive values for 4, 6, 8, and 12 SoC bands (+565, +533, +576, and +561 EUR). Throughput is reported negative except for the 2 MWh case (+19 EUR). These values describe the original runs and have not been independently reproduced.

VII. DISCUSSION

A. Interpretation of the market-position mechanism

Throughput and daily-depth costs depend on energy processed or the SoC excursion. ReLCCA also prices energy retained in each SoC band. Under an upward-reserve commitment, that occupancy cost can alter the marginal value of stored energy and the selected day-ahead position. The original 1 May decomposition reports a much larger difference in day-ahead energy revenue than in direct aging cost. Because the available reconstruction does not reproduce that decomposition, the proposed market-position mechanism is plausible within the stated formulation but is not yet empirically confirmed by an independent replication.

B. Scope of the empirical evidence

The evidence does not establish universal superiority of ReLCCA, a universal negative-price threshold, a population-level reversal probability, or a validated cell-level degradation law. Reserve and imbalance prices are proxies, so the reported euro values are simulated wholesale-profit outputs under the stated proxy assumptions, not realized German settlements.

VIII. LIMITATIONS AND REPRODUCIBILITY

Limitations of the data and sample: The original manuscript reports one reversal in 262 complete days, and the panel does not cover September through November. That frequency cannot be generalized to a full year. The panel spans 1 December 2025 through 21 August 2026 and contains national system series, not battery telemetry.

Limitations of the market model: aFRR capacity prices are repeated four-hour block means rather than day-specific cleared auction records, and activation trajectories are not simulated. Imbalance prices use a two-price proxy rather than observed reBAP settlements. The VPP is a price taker and omits network constraints and strategic price-making. The event scan is limited to the available panel.

Limitations of the battery and aging model: ReLCCA is an optimization-oriented encoding of published aging relationships, not a cell-level electrochemical model. Its coefficients require chemistry-, temperature-, and history-specific calibration. The calendar-cost annualization is linear in time, while the square-root-time case is only a sensitivity test. The EUR 150/kWh replacement value is a cell-cost assumption, not installed EPC cost. The Xu benchmark represents daily depth and is not rainflow counting. Initial SoC, reserve-sustainment duration, and band-specific cycle coefficients are code assumptions, not manuscript-reported values.

Scenario-count sensitivity (reconstruction, 1 May 2026, MILP): the P1−B2 gaps remain negative at every tested scenario count. With K = 30 they are −98.7 (throughput), −97.2 (Xu), and −102.5 (ReLCCA) EUR; with K = 50, −134.6, −103.2, and −132.7; with K = 100, −269.0, −104.5, and −244.0; and with K = 200, −280.7, −253.1, and −253.5. The reconstruction therefore does not reproduce the original positive ReLCCA result at any tested K, so the non-reproduction is a property of the model and input snapshot rather than an artifact of the K = 30 scenario sample. Gap magnitudes vary across K because the MILP dispatch schedule itself changes with the scenario sample; the controlled diagnostic is sign stability, and it holds for all three aging laws.

Limitations of reproducibility: The original manuscript reports Python/Pyomo with HiGHS and seed 20260821. The original source code, original input snapshot, and exact run configuration were not supplied. The accompanying reconstruction uses SciPy's MILP interface to HiGHS; it includes assumptions for initial SoC, reserve-sustainment duration, and separate charge/discharge gamma coefficients. It uses the archived public panel but does not reproduce the original 1 May ranking or negative-price scan. The reconstruction outputs are archived separately and must not replace Tables III–VI. Exact replication requires the original source, original input snapshot, and complete parameter set. Reconstruction diagnostics (labelled reconstruction): across the twenty negative-price scan days, the paired ReLCCA−throughput wholesale-profit difference has mean −3.81 EUR (Wilcoxon signed-rank p = 0.330; block-bootstrap 95% CI [−9.78, +1.24] EUR), and the

ReLCCA differential is the largest of the three aging laws on only 6 of the 20 days; the reconstruction exhibits no systematic ReLCCA advantage on the scan. Replication status: the original manuscript's numerical tables are preserved as reported evidence, while the available independent reconstruction fails to reproduce the key 1 May ranking and the historical event scan. The paper's quantitative conclusions must therefore remain provisional pending access to the original model, input snapshot, and complete parameter set.

IX. CONCLUSION

This study examines whether SoC-occupancy aging changes a BESS energy–reserve decision under stressed prices. The manuscript's original runs report a 1 May 2026 reversal between the throughput/Xu and ReLCCA formulations, with most of the reported profit difference arising from the day-ahead position rather than direct aging cost. The available reconstruction does not reproduce that ranking, so the original numerical conclusion remains provisional until the original model and parameters are available. The analysis supports a narrower motivation for explicit occupancy costs; it does not establish a universal degradation law or realized German settlement outcome.

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